

PicoCTF 2026 – Password Profiler

Password Profiler



General Skills **Easy** 100 pts 4,240 solves

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We intercepted a suspicious file from a system, but instead of the password itself, it only contains its SHA-1 hash. Using OSINT techniques, you are provided with personal details about the target. Your task is to leverage this information to generate a custom password list and recover the original password by matching its hash.

Download the following files:

userinfo: Contains the personal details.

hash: Contains the SHA-1 hash of the password.

check_password: Script to test passwords against the hash.

In this challenge, we're asked to build a custom wordlist to crack a password hash

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Building a Custom Wordlist

```
First Name: Alice  
Surname: Johnson  
Nickname: AJ  
Birthdate: 15-07-1990  
Partner's Name: Bob  
Child's Name: Charlie
```

These are the details about the user that we're given, and we can use a tool called CUPP to create a large list of potential passwords based on these details

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The CUPP Tool

```
cupp.py!  
  
# Common  
# User  
# Passwords  
# Profiler  
  
[ Muris Kurgas | j0rgan@remote-exploit.org ]  
[ Mebus | https://github.com/Mebus/ ]  
  
[+] Insert the information about the victim to make a dictionary  
[+] If you don't know all the info, just hit enter when asked! ;)  
  
> First Name: Alice  
> Surname: Johnson  
> Nickname: 
```

The CUPP (Common User Passwords Profiler) tool can create a large list of passwords based on user information passed into it

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Cracking the Hash

```
└─$ john --wordlist=./alice.txt --format=raw-sha1 ./hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (Raw-SHA1 [SHA1 256/256 AVX2 8x])
Warning: no OpenMP support for this hash type, consider --fork=4
Press 'q' or Ctrl-C to abort, almost any other key for status
AJ_15781194    (?)
```

With the custom password list, we can use the John the Ripper hash cracking tool and try to find a match to the hash

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Obtaining the Flag

```
└─$ python check_password.py  
Password found: picoCTF{4j_15941941}
```

Once a match is found, we can submit it to the checker script found in the challenge description to obtain the flag